

ALI RAZA

SOFTWARE DEVELOPER | AI & SYSTEMS ENGINEERING

Python • AI / LLMs • Linux • Full-Stack Development • Automation

PROGRAMMING

Python • JavaScript • HTML • CSS • Bash

AI & LLM ENGINEERING

OpenAI API / SDK • Ollama • OpenRouter • LangChain • Kokoro • Whisper • AI Agents • Tool Integration • Prompt Design • Local AI

WEB DEVELOPMENT

Flask • FastAPI • React • REST APIs • Full-Stack Dev

DATA & MEMORY

SQLite • ChromaDB • Pandas • Vector Retrieval • Relational Data

AUTOMATION

Playwright • PyAutoGUI • ydotool • SearXNG • Cron • FFmpeg • MoviePy • yt-dlp

PROFESSIONAL PROFILE

Software developer and technology entrepreneur with practical experience designing, developing, deploying, and maintaining software across web applications, AI assistants, automation systems, and Linux environments. Core strengths are in Python development, AI/LLM integration, software architecture, automation, APIs, Linux systems, databases, and full-stack web development. Experienced in taking projects from concept through iterative development, debugging, architectural changes, and deployment. Combines technical problem-solving with creative design experience and a strong interest in building autonomous and highly integrated software systems.

WORK EXPERIENCE

Founder & Developer

AUG 2025 – FEB 2026

Gadget Labs

- Founded and managed an experimental technology-focused e-commerce and dropshipping business.
- Developed the company website and supporting technical infrastructure, integrating automated solutions to reduce repetitive business processes.
- Managed technical implementation from concept through operation, working independently across development, design, branding, and business operations.

Freelance Web Developer

MAY 2024 – JUN 2024

BWA-PC Builds

- Designed and developed an e-commerce website structure and visual interface using HTML5, CSS3, and Figma.
- Translated business requirements into a functional website and iterated independently against practical client requirements.

MAJOR PROJECTS

CIEL (Central Intelligence and Execution Layer) **BETA TESTING**

Tech: Python, OpenAI API, Ollama, Kokoro, ChromaDB, SQLite, Playwright, PyAutoGUI, Arch Linux, D-Bus

Locally focused AI assistant and execution layer combining conversational AI with computer interaction, automation, persistent memory, autonomous workflows. and external services.

LINUX & SYSTEMS

[Arch Linux](#) • [Linux](#) •
[Hyprland](#) • [archiso](#) •
[mkinitcpio](#) • [systemd](#) • [D-Bus](#) •
[Desktop Auto](#)

DEVELOPMENT TOOLS

[Git](#) • [GitHub](#) • [VS Code](#) •
[Codex](#)

DESIGN

[Figma](#) • [Photoshop](#) •
[Illustrator](#) • [Premiere Pro](#) •
[DaVinci Resolve](#) • [UI/UX](#)

LANGUAGES

- **Urdu** — Native
- **English** — Fluent
- **Spanish** — Fluent / Near-native

PROBLEM SOLVING & ENG

- Analytical & Systems thinking
- Modular troubleshooting
- Software architecture
- Iterative development
- System integration

WORKING STYLE

- Self-directed learning
- Project ownership & Initiative
- Independent decision-making
- Adaptability & Detail-oriented

INTERESTS

- AI & Systems Architecture
- Linux & Open Source
- Emerging Technology
- Digital Design & Entrepreneurship

SELECTED PROJECTS

LifeOS **DEPLOYED**

Tech: Python, Flask, SQLite3, HTML, CSS, JavaScript

A full-stack personal management platform centralising tasks, habits, goals, projects, journaling, and other areas of personal organisation.

- Designed a Python/Flask backend with a database-backed application structure and a standard frontend.
- Developed features iteratively, restructuring components when the original architecture showed limitations.
- Built API integration into the platform so external applications (e.g., CIEL) can interact with LifeOS data.
- Worked across frontend, backend, database, persistent state, and inter-application communication.

AutoDesign Markup Language (ADML) **IN DEVELOPMENT**

Tech: Python, Regex, python-pptx

A custom markup language giving AI systems a structured way to describe professional visual designs.

- Designed a custom syntax with a parser and compiler pipeline to interpret ADML instructions.
- Explores structured intermediate representations to make AI-generated designs predictable and controllable.

Portable Compute Protocol (PCP) **IN DEVELOPMENT**

Tech: Arch Linux, archiso, Bash, mkinitcpio

An experimental portable computing environment based around an Arch Linux system capable of operating primarily from RAM.

- Investigated Linux image-generation and boot infrastructure, RAM-based execution, and portability.
- Explored the interaction between hardware, operating systems, storage, and boot infrastructure.

EDUCATION

The Sixth Form Bolton

SEP 2024 – 2026

- A Level Computer Science
- A Level Graphics
- BTEC Certificate in ICT

Essa Academy

Graduated 2024

- 8 GCSEs at Grade 4 or above, including Computer Science, Graphics, Mathematics, Science, English Language, and English Literature.